A LOCAL HOMOMORPHISM IS A COLIMIT OF LOCAL HOMOMORPHISMS BETWEEN ESSENTIALLY FINITELY GENERATED LOCAL RINGS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Finite subsets as an index for a directed colimit. Used to define the index set in the counted proof (lines 32456-32460). Provides the directed set over which the approximations are taken.
- Localisation at the maximal ideal to keep the rings local. Used in the construction of each approximation (the proof localises the finitely generated pieces). Makes each approximating ring local and each map a local homomorphism, which is required by the statement.

Lemma 0.1. *Suppose $R \to S$ is a local homomorphism of local rings. There exists a directed set $(\Lambda, \leq)$, and a system of local homomorphisms $R_\lambda \to S_\lambda$ of local rings such that*

(1) *The colimit of the system $R_\lambda \to S_\lambda$ is equal to $R \to S$.*
(2) *Each R_λ is essentially of finite type over $\mathbf{Z}$.*
(3) *Each S_λ is essentially of finite type over R_λ.*

Proof. Denote $\varphi : R \to S$ the ring map. Let $\mathfrak{m} \subset R$ be the maximal ideal of R and let $\mathfrak{n} \subset S$ be the maximal ideal of S. Let

$$\Lambda = \{(A, B) \mid A \subset R, B \subset S, \#A < \infty, \#B < \infty, \varphi(A) \subset B\}.$$

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As partial ordering we take the inclusion relation. For each $\lambda = (A, B) \in \Lambda$ we let R'_λ be the sub $\mathbf{Z}$-algebra generated by $a \in A$, and we let S'_λ be the sub $\mathbf{Z}$-algebra generated by b, $b \in B$. Let R_λ be the localization of R'_λ at the prime ideal $R'_\lambda \cap \mathfrak{m}$ and let S_λ be the localization of S'_λ at the prime ideal $S'_\lambda \cap \mathfrak{n}$. In a picture

$$\begin{array}{ccccccc} B & \longrightarrow & S'_\lambda & \longrightarrow & S_\lambda & \longrightarrow & S \\ \uparrow & & \uparrow & & \uparrow & & \uparrow \\ A & \longrightarrow & R'_\lambda & \longrightarrow & R_\lambda & \longrightarrow & R \end{array} .$$

The transition maps are clear. We leave the proofs of the other assertions to the reader. $\square$